UC SAN DIEGO / ECE 251C

12

Frequency decomposition and sparsity

Executive summary

The DWT separates a signal into a coarse approximation and details at successive scales. We first interpret these scales as octave frequency bands and compare the time–frequency resolution of wavelets with that of Fourier representations. We then explain why localization and vanishing moments produce sparse coefficients for signals that are locally simple. Haar cancels constant pieces, while higher-order wavelets cancel polynomial pieces. Finally, we derive the tradeoff between vanishing moments and support length and show how the Daubechies construction attains the minimum filter length.

12.1 Reading the DWT as a frequency decomposition

The preceding lectures established how to construct a wavelet transform and recover a signal from its coefficients. Perfect reconstruction ensures that no information is lost. To understand why the new coordinates are useful, we must also understand what an individual coefficient measures. We begin with the frequency information captured by each branch of the analysis bank. We will then connect the scale of a wavelet to its localization and its ability to cancel simple behavior away from transitions.

At the finest available resolution I, write

$$f_I(t) = \sum_n a_I[n]\phi_{I,n}(t), \qquad \phi_{i,n}(t) = 2^{i/2}\phi(2^i t - n).$$

Here $a_i[n] = \langle f, \phi_{i,n}\rangle_{L^2}$, and the wavelet coefficients are $d_i[n] = \langle f, \psi_{i,n}\rangle_{L^2}$, with $\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n)$. The function f_I is the projection of f onto V_I. One analysis stage computes

$$a_{i-1}[n] = (h_0 * a_i)[2n], \qquad d_{i-1}[n] = (h_1 * a_i)[2n]. \tag{12.1}$$

The low-pass output describes V_{i-1}; the detail output describes W_{i-1}, where $V_i = V_{i-1} \oplus W_{i-1}$. Only the approximation branch is decomposed again in the usual DWT.

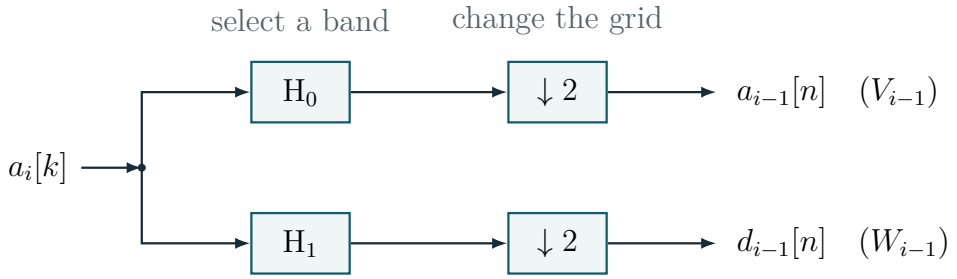


Figure 12.1: One analysis stage separates approximation and detail. Downsampling changes the frequency coordinates of both outputs.

12.2 What downsampling does to the frequency axis

A discrete angular frequency is measured in radians per sample. Since downsampling changes the sample spacing, it also changes the numerical label assigned to a fixed oscillation. To interpret the frequency bands in a multilevel DWT, we must therefore distinguish the frequency variable of the original input from that of each downsampled branch.

Use the DTFT convention

$$\widehat{x}(\omega) = X(\mathrm{e}^{\mathrm{j}\omega}) = \sum_{n\in\mathbb{Z}} x[n]\mathrm{e}^{-\mathrm{j}\omega n}.$$

Suppose $y[n] = x[2n]$. We use θ for the output's frequency variable. Starting with finitely supported sequences makes every rearrangement finite:

$$\begin{aligned}
&\frac{1}{2}\left[\widehat{x}(\theta/2) + \widehat{x}(\theta/2+\pi)\right] \\
&\quad = \sum_k x[k]\mathrm{e}^{-\mathrm{j}\theta k/2}\frac{1+\mathrm{e}^{-\mathrm{j}\pi k}}{2} \\
&\quad = \sum_{k \text{ even}} x[k]\mathrm{e}^{-\mathrm{j}\theta k/2} \\
&\quad = \sum_n x[2n]\mathrm{e}^{-\mathrm{j}\theta n} = \widehat{y}(\theta).
\end{aligned}$$

The factor $(1+(-1)^k)/2$ is one on even indices and zero on odd indices. Thus

$$\boxed{\widehat{y}(\theta) = \frac{1}{2}\left[\widehat{x}(\theta/2) + \widehat{x}(\theta/2+\pi)\right].} \tag{12.2}$$

The second argument can also be written $(\theta - 2\pi)/2$: the two versions differ by one DTFT period. The identity extends to ℓ^2 sequences as an equality of L^2 frequency responses, by finite truncation and Plancherel.

Frequency scaling for one oscillation. If $x[n] = \mathrm{e}^{\mathrm{j}\omega n}$, then

$$y[n] = \mathrm{e}^{\mathrm{j}(2\omega)n}.$$

Hence the output label is $\theta = 2\omega$ modulo 2π. In particular,

$$\omega = \frac{\pi}{4} \;\mapsto\; \theta = \frac{\pi}{2}, \qquad \omega = \frac{3\pi}{4} \;\mapsto\; \theta = -\frac{\pi}{2}.$$

These exponentials are diagnostic inputs, not finite-energy sequences. The second example shows the frequency folding that occurs in the high-pass branch.

If filtering precedes downsampling, replace $\hat{x}$ in (12.2) by $\hat{h}\,\hat{x}$. The filter selects a band *before* the frequency axis is expanded and wrapped.

12.3 An idealized first split

For a full-band input, consider ideal analysis magnitudes

$$|\hat{h}_0(\omega)| = \sqrt{2}\,\mathbf{1}_{\{|\omega|<\pi/2\}}, \qquad |\hat{h}_1(\omega)| = \sqrt{2}\,\mathbf{1}_{\{\pi/2<|\omega|<\pi\}},$$

extended periodically. Endpoint values do not affect L^2 statements. These are idealized bands, not FIR filters; phases can be chosen to obtain the Shannon orthogonal bank. We are tracking support here, not prescribing those phases.

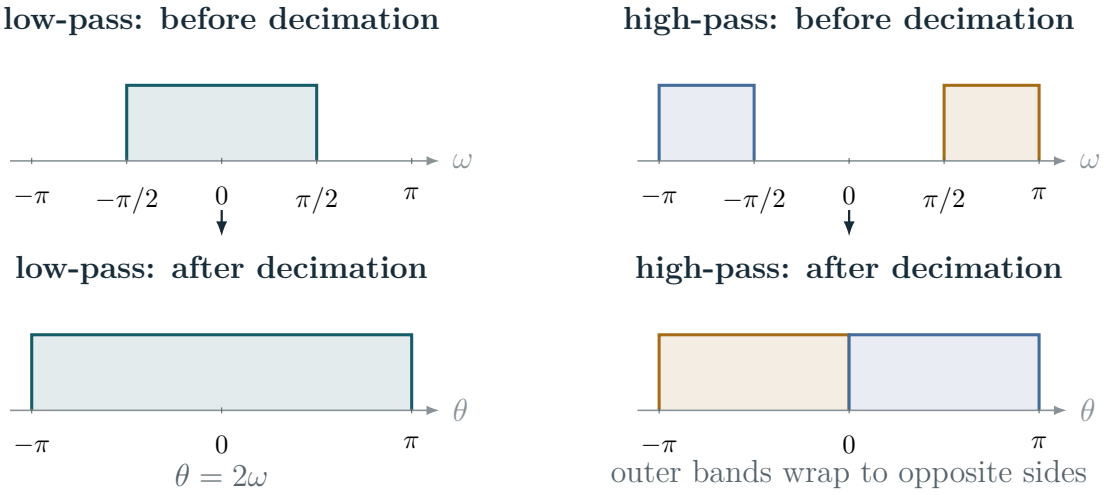


Figure 12.2: Ideal band locations before and after downsampling. Heights indicate occupied bands only, not amplitudes. The colors track which original high-frequency half-band goes to each output half.

In the low-pass branch, only the central half-band survives. For $-\pi < \theta < \pi$, the shifted term in (12.2) vanishes, so

$$\hat{a}_{I-1}(\theta) = \tfrac{1}{2}\hat{h}_0(\theta/2)\hat{a}_I(\theta/2).$$

The retained interval $[-\pi/2, \pi/2]$ is mapped onto the full output interval $[-\pi, \pi]$. Notice also the factor $1/2$ in the downsampling formula. The diagram records

which frequencies are present; its vertical heights do not specify the output amplitudes.

The detail branch retains the two outer half-bands. Decimation maps them into the principal interval of the output DTFT, with the positive and negative parts wrapping to opposite sides. The resulting coefficient sequence still describes high-frequency information in the original signal. Its own DTFT occupies $[-\pi, \pi]$ because it is measured on a different sampling grid.

12.4 Repeated low-pass splitting produces octaves

After one low-pass stage, the branch frequency is $\theta_1 = 2\omega$ on the retained band. After k low-pass stages it is $\theta_k = 2^k\omega$. The next low-pass filter again retains $|\theta_k| < \pi/2$, now meaning $|\omega| < \pi/2^{k+1}$.

Induction gives the bands below, all in the *original* variable ω:

$$\boxed{\begin{aligned} V_{I-k}: &\quad |\omega| \le \frac{\pi}{2^k}, \\ W_{I-k}: &\quad \frac{\pi}{2^k} \le |\omega| \le \frac{\pi}{2^{k-1}}, \qquad k \ge 1. \end{aligned}} \tag{12.3}$$

Here the subspace labels refer to the reconstructed approximation and detail functions, while ω is measured on the finest coefficient grid. The band interpretation is exact for the ideal Shannon construction. For general wavelets it describes nominal bands rather than disjoint Fourier supports; the orthogonal decomposition of the spaces remains exact.

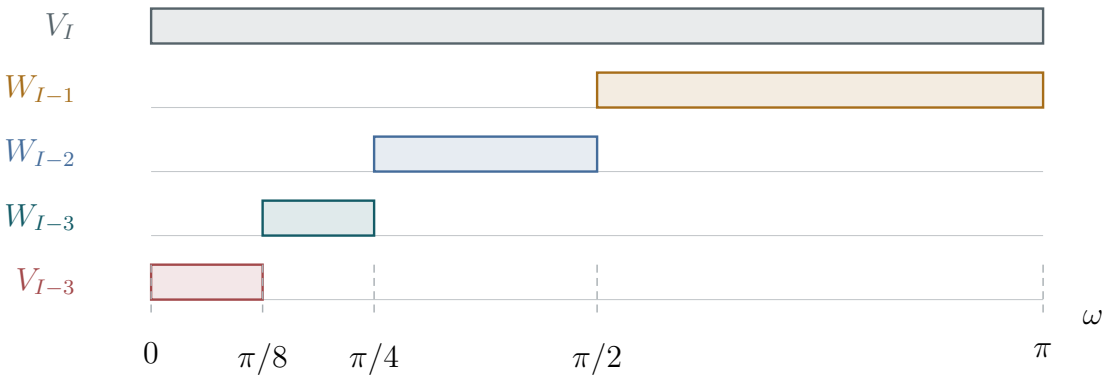


Figure 12.3: Three levels on one frequency axis, showing positive frequencies. The negative-frequency bands are their reflections. Bandwidth halves as the decomposition moves down the low-pass branch.

The orthogonal identity itself is exact for every orthogonal MRA:

$$V_I = V_{I-3} \oplus W_{I-3} \oplus W_{I-2} \oplus W_{I-1}.$$

Each positive detail band has upper endpoint twice its lower endpoint: it has width one on the $\log_2 \omega$ axis. This is the *octave* structure, named for the frequency ratio of two in a musical octave. The logarithmic organization also has an analogue in the response characteristics of the human ear. The comparison concerns the organization of frequency information; it does not identify hearing with a dyadic DWT.

Discrete and continuous frequencies. If Ω denotes continuous-domain angular frequency, direct transformation of f_I gives

$$\widehat{f}_I(\Omega) = 2^{-I/2}\widehat{\phi}(2^{-I}\Omega)\widehat{a}_I(2^{-I}\Omega). \tag{12.4}$$

Thus $\omega = 2^{-I}\Omega$ on the finest coefficient grid. The extra factor $\widehat{\phi}$ matters: an arbitrary V_I is not a brick-wall bandlimited space.

12.5 Practical filters have overlapping transition bands

The sharp bands organize the DWT, but are not the exact spectra of compactly supported wavelets. Haar has normalized analysis powers

$$\rho_0(\omega) \coloneqq \frac{|\widehat{h}_0(\omega)|^2}{2} = \cos^2(\omega/2), \qquad \rho_1(\omega) \coloneqq \frac{|\widehat{h}_1(\omega)|^2}{2} = \sin^2(\omega/2).$$

They satisfy $\rho_0 + \rho_1 = 1$, but neither is a rectangular band.

Define the normalized three-level branch profiles by

$$A_3(\omega) = \prod_{r=0}^{2} \rho_0(2^r\omega),$$

$$D_k(\omega) = \rho_1(2^{k-1}\omega) \prod_{r=0}^{k-2} \rho_0(2^r\omega), \qquad k = 1, 2, 3.$$

An empty product equals one. Each deeper filter acts on a downsampled grid. The equivalent depth-k filter on the original grid is

$$B_k(z) = H_1(z^{2^{k-1}}) \prod_{r=0}^{k-2} H_0(z^{2^r}), \qquad D_k(\omega) = 2^{-k}|B_k(\mathrm{e}^{\mathrm{j}\omega})|^2.$$

For example, two filtering stages combine as $\sum_{m,r} h_0[m]h_0[r]x[4n-2m-r]$, giving $H_0(z^2)H_0(z)$ before decimation by four.

Put $A_k = \prod_{r=0}^{k-1} \rho_0(2^r\omega)$ and $A_0 = 1$. Then $D_k = A_{k-1} - A_k$, so

$$A_3 + D_3 + D_2 + D_1 = 1.$$

The telescoping identity explains the complementary shapes in the figure. It does not say that the energy of a decimated output is obtained by multiplying the

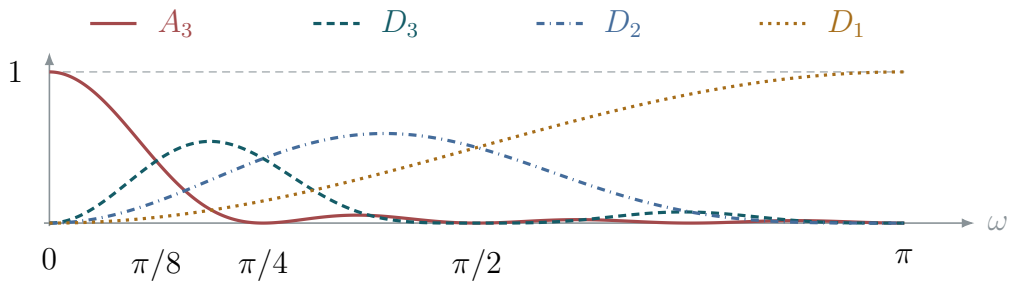


Figure 12.4: Actual normalized Haar branch powers. Transition regions overlap, and deeper bands have sidelobes. The four profiles sum to one, but are not disjoint frequency windows.

input spectrum pointwise by one of these profiles: downsampling also introduces the alias terms derived above. In particular, overlapping transition bands are compatible with an orthogonal filter bank.

12.6 Four ways to allocate time–frequency resolution

The frequency decomposition also has a spatial interpretation. Fine-scale wavelets are narrow and, under the unit-energy normalization, tall. They capture localized high-frequency information, such as an edge in an image. Coarse-scale wavelets are wider and shorter, so they capture changes over longer intervals, such as the slowly varying background of an image. Time–frequency diagrams make this tradeoff visible and let us compare it with other signal representations.

A coefficient is an inner product with an analyzing function q:

$$c_q = \langle f, q \rangle_{L^2} = \int_{\mathbb{R}} f(t) q^*(t)\, \mathrm{d}t = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\Omega) \hat{q}(\Omega)^*\, \mathrm{d}\Omega.$$

The coefficient depends most strongly on the part of the signal where the analyzing function is concentrated. Parseval gives the same interpretation in frequency. A time–frequency rectangle summarizes these two concentrations. It is not an exact joint support: a nonzero function cannot be confined to bounded intervals in both domains.

For an N-sample signal, the unit impulse $e_n[k] = \delta[k-n]$ has unitary DFT

$$(\mathrm{F} e_n)[m] = N^{-1/2} \mathrm{e}^{-\mathrm{j}2\pi mn/N}.$$

It occupies one time sample and all frequency bins. Conversely, the Fourier vector $N^{-1/2}\mathrm{e}^{\mathrm{j}2\pi mk/N}$ has constant magnitude over the whole record and occupies one DFT bin. These give the vertical and horizontal strips.

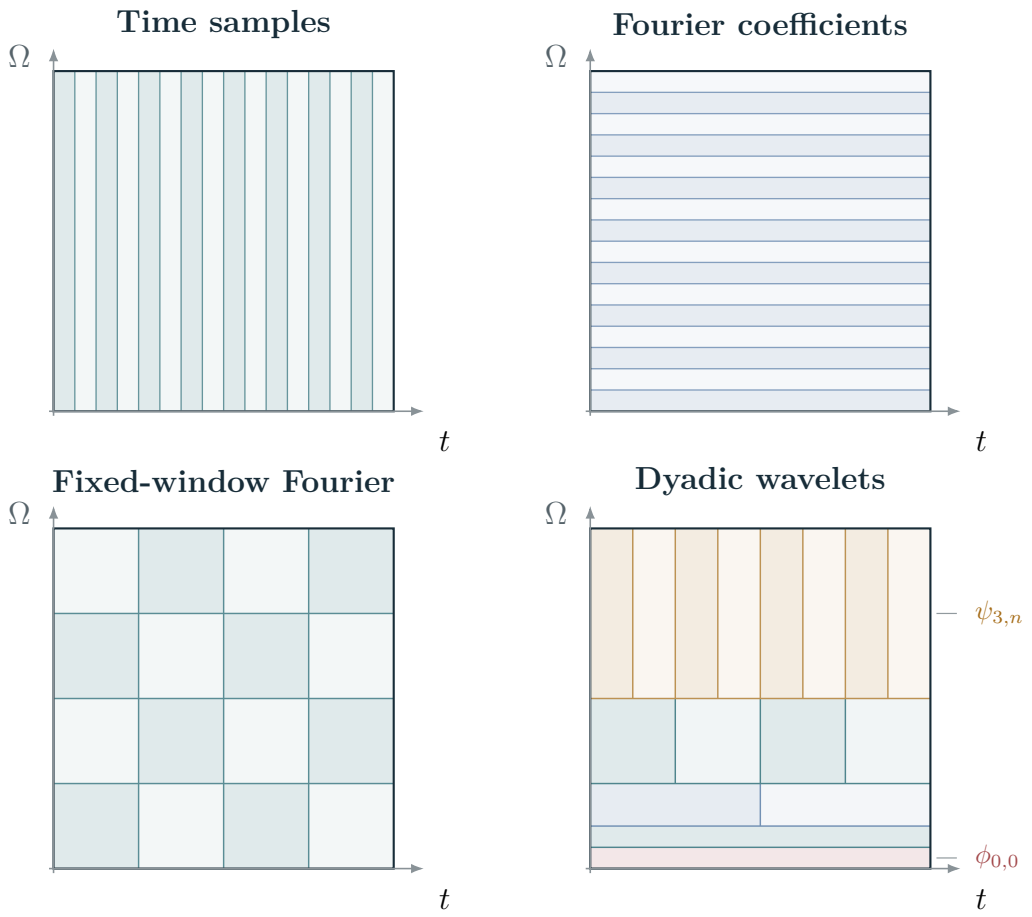


Figure 12.5: Each panel has 16 equal-area nominal coefficient cells, not exact joint supports. The wavelet panel retains one scaling cell, followed by 1, 2, 4, and 8 detail cells as frequency increases.

Fixed windows give fixed resolution. For a unit-norm window w, define

$$q_{u,\xi}(t) = w(t-u)\mathrm{e}^{\mathrm{j}\xi t}, \qquad V_w f(u,\xi) = \langle f, q_{u,\xi}\rangle_{L^2}.$$

This is the short-time Fourier transform (STFT); $|V_w f|^2$ is the spectrogram. Substituting $s = t - u$ gives

$$\widehat{q}_{u,\xi}(\Omega) = \mathrm{e}^{-\mathrm{j}(\Omega-\xi)u}\widehat{w}(\Omega-\xi).$$

Translation shifts the time concentration, and modulation shifts the frequency concentration. Neither operation changes the corresponding width. Thus a fixed window uses the same resolution tradeoff throughout the time–frequency plane. Sampling the STFT more densely changes the number of coefficients, not the localization of the individual analyzing functions.

Wavelet dilation changes both widths. For $\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n)$, put $s = 2^i t - n$:

$$\begin{aligned}\widehat{\psi}_{i,n}(\Omega) &= 2^{i/2}\int_{\mathbb{R}} \psi(2^i t - n)\mathrm{e}^{-\mathrm{j}\Omega t}\,\mathrm{d}t \\ &= 2^{-i/2}\int_{\mathbb{R}} \psi(s)\mathrm{e}^{-\mathrm{j}2^{-i}\Omega(s+n)}\,\mathrm{d}s \\ &= 2^{-i/2}\mathrm{e}^{-\mathrm{j}2^{-i}n\Omega}\widehat{\psi}(2^{-i}\Omega).\end{aligned}$$

An interval containing a fixed fraction of the time energy is compressed by 2^{-i}; the corresponding frequency region is expanded by 2^i. Thus

$$\boxed{\Delta t_i = 2^{-i}\Delta t_\psi, \quad \Delta\Omega_i = 2^i\Delta\Omega_\psi, \quad \Delta t_i\Delta\Omega_i = \Delta t_\psi\Delta\Omega_\psi.} \tag{12.5}$$

The translation step is also 2^{-i}: neighboring coefficients at a finer scale interrogate more closely spaced locations.

Nominal cells versus actual localization. A precise convention for the positive-frequency diagram is

$$[2^{-i}n, 2^{-i}(n+1)) \times [2^i\pi, 2^{i+1}\pi),$$

with a bottom scaling row $[n, n+1) \times [0, \pi)$. Every cell has area π. These cells partition coefficient locations; actual wavelets can extend beyond their cell. Real wavelets have symmetric positive- and negative-frequency energy, so the drawing shows only one side of their spectrum.

For a unit-norm atom q, suppose $\int_{\mathbb{R}\setminus T} |q(t)|^2\,\mathrm{d}t \le \varepsilon$. Then Cauchy–Schwarz gives

$$\left|\langle f, q\rangle - \int_T f(t)q^*(t)\,\mathrm{d}t\right| \le \sqrt{\varepsilon}\,\|f\|_{L^2}.$$

Thus little atom energy outside T limits the contribution from that region. Parseval gives the analogous statement for a frequency-concentration region.

12.7 The uncertainty principle

The reciprocal widths in a wavelet family follow from dilation. A related question is how tightly any analyzing function can be concentrated in both time and frequency. The uncertainty principle answers this question when concentration is measured by variance.

For a unit-norm atom q, suppose the following means and variances exist:

$$\tau = \int t|q(t)|^2\,\mathrm{d}t, \qquad \nu = \frac{1}{2\pi}\int \Omega|\widehat{q}(\Omega)|^2\,\mathrm{d}\Omega,$$

$$\sigma_t^2 = \int (t-\tau)^2 |q(t)|^2 \,\mathrm{d}t, \quad \sigma_\Omega^2 = \frac{1}{2\pi}\int (\Omega-\nu)^2 |\hat{q}(\Omega)|^2 \,\mathrm{d}\Omega.$$

All integrals here are over $\mathbb{R}$. The uncertainty inequality is

$$\boxed{\sigma_t \sigma_\Omega \geq \tfrac{1}{2}.} \tag{12.6}$$

Proof for smooth, rapidly decaying atoms. Translation and modulation let us assume $\tau = \nu = 0$ without changing the variances. Integration by parts, followed by Cauchy–Schwarz, gives

$$\begin{aligned} 1 &= \int |q(t)|^2 \,\mathrm{d}t = -\int t \frac{\mathrm{d}}{\mathrm{d}t}|q(t)|^2 \,\mathrm{d}t \\ &= -2\,\mathrm{Re}\int t q(t)^* q'(t)\,\mathrm{d}t \leq 2\|tq\|_{L^2}\|q'\|_{L^2}. \end{aligned}$$

The boundary term is zero by decay. Plancherel and $\widehat{q'}(\Omega) = \mathrm{j}\Omega\hat{q}(\Omega)$ give $\|tq\|_{L^2} = \sigma_t$ and $\|q'\|_{L^2} = \sigma_\Omega$, proving the bound. □

The finite-variance version follows by approximating the atom by smooth, rapidly decaying functions in the norms appearing in the proof. The conclusion is a lower bound on the product of the two spreads. It does not require different atoms to have the same uncertainty product. Within a fixed wavelet family, the constant-area relation instead follows directly from the reciprocal dilation factors in (12.5).

Haar requires a different notion of width. Direct integration yields

$$\hat{\psi}(\Omega) = \frac{(1-\mathrm{e}^{-\mathrm{j}\Omega/2})^2}{\mathrm{j}\Omega}, \qquad \Omega^2|\hat{\psi}(\Omega)|^2 = 16\sin^4(\Omega/4).$$

At $\Omega = 0$, the Fourier formula is interpreted by continuity. The function $16\sin^4(\Omega/4)$ has an infinite integral, so Haar does not have finite frequency variance. Its time–frequency diagrams must therefore be read as nominal band allocations or energy-concentration regions, not as finite-variance boxes. The exact dilation formula still explains their multiscale geometry.

12.8 Haar coefficients vanish on constant pieces

The frequency-band picture describes how wavelets organize different resolutions. Sparsity requires another property: cancellation. For Haar this is especially simple because each wavelet compares the integrals over two neighboring half-intervals. Recall

$$\phi(t) = \mathbf{1}_{[0,1)}(t), \qquad \psi(t) = \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).$$

The Haar wavelet has zero mean:

$$\int_{\mathbb{R}} \psi(t)\,\mathrm{d}t = \frac{1}{2} - \frac{1}{2} = 0.$$

The positive and negative halves therefore cancel whenever the signal is constant over the wavelet's support.

Let $I_{i,n} = [2^{-i}n, 2^{-i}(n+1))$, with left and right halves $I_{i,n}^{\mathrm{L}}$ and $I_{i,n}^{\mathrm{R}}$. Then

$$\boxed{d_i[n] = 2^{i/2}\left[\int_{I_{i,n}^{\mathrm{L}}} f(t)\,\mathrm{d}t - \int_{I_{i,n}^{\mathrm{R}}} f(t)\,\mathrm{d}t\right].} \tag{12.7}$$

If $f = c$ on $I_{i,n}$, both integrals equal $c2^{-i-1}$, so $d_i[n] = 0$.

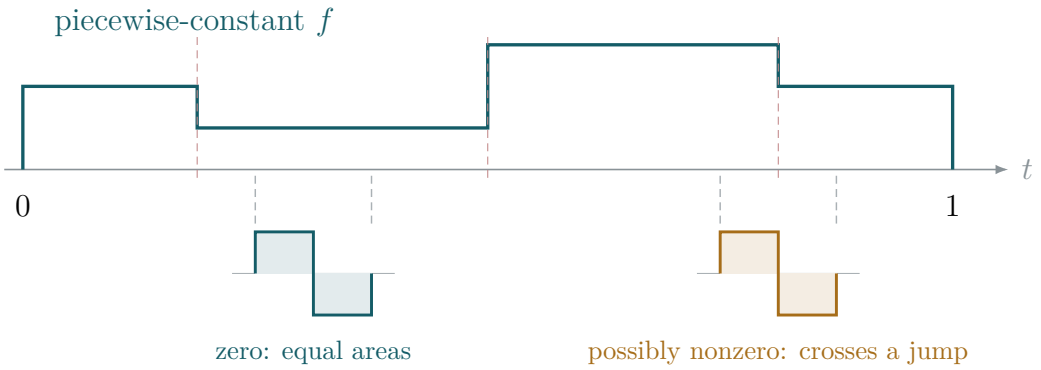


Figure 12.6: A Haar atom inside a flat piece has zero coefficient. Only supports that straddle transitions can contribute. The small atoms are drawn with a common illustrative height; their actual normalization is $2^{i/2}$.

For a piecewise-constant signal, a nonzero Haar detail can occur only when the *interior* of its supporting interval contains a jump. A jump at a cell boundary does not create a detail in either adjacent cell, since both cells are individually constant. A wavelet crossing several jumps can also have zero coefficient through cancellation; crossing a jump is necessary, not sufficient, for a nonzero coefficient.

This argument does not say that a piecewise-constant signal has only low-frequency content. A jump produces Fourier coefficients at arbitrarily high frequencies. The advantage of Haar is that, at each scale, only wavelets whose supports cross a transition can have nonzero detail coefficients. The representation records the location of the variation as well as its scale.

The coefficient of one jump. On $I_{i,n} = [a, a+\ell)$, where $\ell = 2^{-i}$, suppose

$$f(t) = c + A\mathbf{1}_{[\tau,\infty)}(t), \qquad \tau = a + v\ell, \quad 0 < v < 1.$$

Here A is the signed jump size. The constant c contributes zero. If $0 < v \le 1/2$, the jump contributes on a length $(1/2 - v)\ell$ in the left half and $\ell/2$ in the right

half. If $1/2 \le v < 1$, it contributes only on a length $(1-v)\ell$ in the right half. Hence

$$d_i[n] = -A2^{-i/2} \begin{cases} v, & 0 \le v \le 1/2, \\ 1-v, & 1/2 \le v \le 1. \end{cases} \tag{12.8}$$

In particular, $|d_i[n]| \le |A|2^{-i/2}/2$. The negative sign for an upward jump follows from our convention that the right half of ψ is negative.

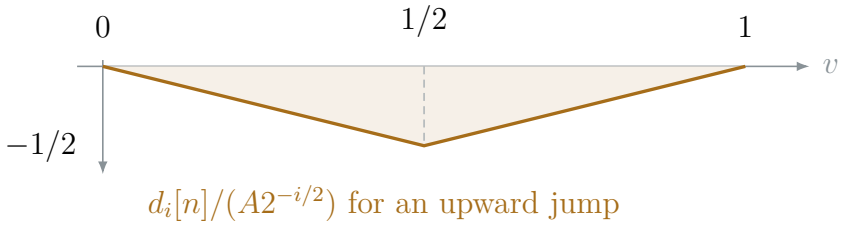


Figure 12.7: The coefficient depends on the jump location within the cell. It is largest in magnitude at the midpoint and vanishes at either boundary.

How many coefficients can be nonzero? Suppose f is piecewise constant on $[0,1]$ with S interior jumps and is zero outside this interval. At scale $i \ge 0$, the dyadic cells partition $[0,1]$. Each jump lies in at most one cell interior, so

$$\#\{n : d_i[n] \ne 0\} \le \min(S, 2^i).$$

At resolutions $0 \le i < I$, at most SI detail coefficients can be nonzero, out of $2^I - 1$ possible details.

The coarse Haar coefficient is just the average:

$$a_0[0] = \int_0^1 f(t)\,\mathrm{d}t, \qquad a_0[n] = 0 \quad (n \ne 0).$$

The full expansion is

$$f = a_0[0]\phi + \sum_{i=0}^{\infty} \sum_{n=0}^{2^i-1} d_i[n]\psi_{i,n} \quad \text{in } L^2.$$

A jump at a non-dyadic location can produce nonzero coefficients at arbitrarily fine scales. Thus sparsity at each scale does not generally mean that the whole infinite expansion has finitely many terms. At a fixed finest resolution, however, the count grows at most linearly with the number of scales rather than with the number of available detail coefficients.

12.9 A concrete sparse expansion

To put numbers on the piecewise-constant picture, take

$$f(t) = 2\mathbf{1}_{[0,1)}(t) - \mathbf{1}_{[3/16,1/2)}(t) + \mathbf{1}_{[1/2,13/16)}(t).$$

Its successive heights on $[0,1]$ are $2,1,3,2$, with jumps at $3/16$, $1/2$, and $13/16$; it is zero outside $[0,1]$. At resolution $I=6$, all 64 scaling coefficients are nonzero. On a cell where $f=c$, their normalization is

$$a_6[n] = \int f(t)\, 8\mathbf{1}_{I_{6,n}}(t)\, \mathrm{d}t = c/8.$$

The Haar recursion is $a_{i-1}[n] = (a_i[2n] + a_i[2n+1])/\sqrt{2}$ and $d_{i-1}[n] = (a_i[2n] - a_i[2n+1])/\sqrt{2}$. Applying it gives

sequence	nonzero indices	values
a_0	0	2
d_0	0	$-5/8$
d_1	$0,1$	$3\sqrt{2}/16$ at each index
d_2	$0,3$	$1/8$ at each index
d_3	$1,6$	$\sqrt{2}/8$ at each index
d_4, d_5	none	0

For example, $d_3[1] = 2^{3/2}(2/16 - 1/16) = \sqrt{2}/8$.

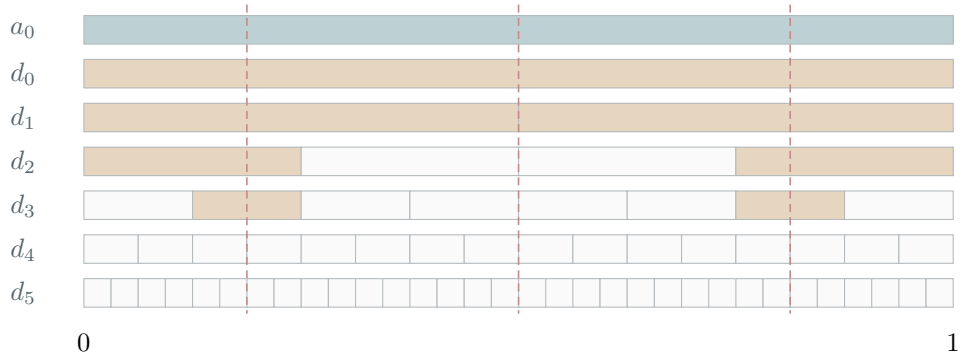


Figure 12.8: Filled cells mark nonzero coefficients, with rows ordered from coarse to fine. Dashed lines mark the three jumps. The finest two rows are entirely zero; this is a coefficient-location display, not a frequency-axis plot.

Only eight of the 64 coefficients are nonzero, including the coarse average. This is an exact change of representation, not a thresholded approximation. The jump locations align with the resolution-four grid, so $f \in V_4$ and every detail at resolution $i \geq 4$ vanishes. As a check, Parseval gives

$$\|f\|_{L^2}^2 = \frac{37}{8} = 2^2 + \left(\frac{5}{8}\right)^2 + 2\left(\frac{3\sqrt{2}}{16}\right)^2 + 2\left(\frac{1}{8}\right)^2 + 2\left(\frac{\sqrt{2}}{8}\right)^2.$$

12.10 Vanishing moments cancel polynomial behavior

Haar cancels the local constant part of a signal. Many signals encountered in applications are approximately piecewise polynomial: between transitions, a low-degree polynomial describes their behavior reasonably well. On such an interval, a polynomial of higher degree can be a better approximation than a constant. This motivates wavelets that cancel a larger collection of polynomial terms. The relevant conditions are the vanishing moments; the Daubechies, Symmlet, and spline wavelet families provide examples with higher cancellation orders.

Definition 12.1 (Vanishing moments)**.** A compactly supported wavelet $\psi \in L^2(\mathbb{R})$ has *at least p vanishing moments* if

$$\int_{\mathbb{R}} t^m \psi(t)\,\mathrm{d}t = 0, \qquad m = 0, 1, \ldots, p-1. \tag{12.9}$$

We call this a wavelet *having p vanishing moments*, with "p" meaning *at least* p unless stated otherwise. Compact support ensures these integrals exist; for noncompact wavelets, assume the corresponding weighted integrability.

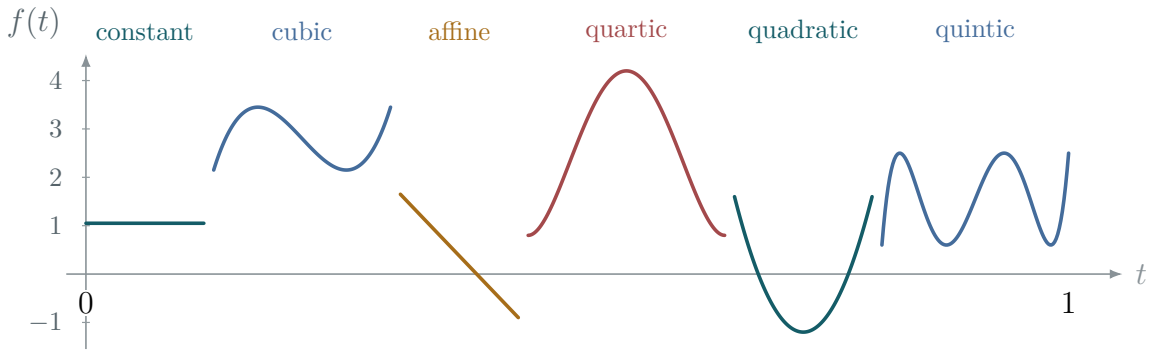


Figure 12.9: Six pieces, of degrees $0, 3, 1, 4, 2, 5$ from left to right. A wavelet with six vanishing moments cancels each piece whenever its entire support lies inside it. Gaps mark transitions; no global smoothness is assumed.

Proposition 12.2 (Local polynomial cancellation)**.** Let ψ have p vanishing moments. If f agrees with a polynomial of degree less than p on the support of $\psi_{i,n}$, then $\langle f, \psi_{i,n} \rangle_{L^2} = 0$.

Proof. It suffices to treat $f(t) = t^r$ on that support, with $r < p$. Set $u = 2^i t - n$. Conjugation is in the second argument, so

$$\begin{aligned} \int t^r \psi_{i,n}(t)^*\,\mathrm{d}t &= 2^{-i(r+1/2)} \int (u+n)^r \psi(u)^*\,\mathrm{d}u \\ &= 2^{-i(r+1/2)} \sum_{m=0}^{r} \binom{r}{m} n^{r-m} \left(\int u^m \psi(u)\,\mathrm{d}u \right)^* = 0. \end{aligned}$$

Linearity gives the claim. Only the compact wavelet support is involved; the polynomial need not belong to $L^2(\mathbb{R})$. □

12.11 More vanishing moments require a longer filter

Additional moment conditions constrain the filter coefficients, and these constraints require a sufficiently long filter. We will prove a sharp lower bound in the real, compactly supported, orthonormal dyadic MRA setting. Orthogonality is an essential part of this statement; the bound is not a restriction on every function with vanishing moments or on every biorthogonal construction.

Theorem 12.3 (Daubechies, 1988: minimum filter length)**.** Let a real FIR lowpass filter g_0 be supported in $\{0, \dots, L-1\}$, with both endpoint taps nonzero. Suppose it generates an orthonormal MRA, has $G_0(1) = \sqrt{2}$, and its wavelet has at least $p \geq 1$ vanishing moments. Then

$$\boxed{L \geq 2p.} \tag{12.10}$$

Here L includes internal zero taps. Unit energy and $G_0(1) = \sqrt{2}$ rule out $L = 1$. The last index $N = L - 1$ is odd: an even $N > 0$ would give the forbidden correlation $g_0[N]g_0[0] \neq 0$ at lag N.

Proof. Set $N = L - 1$. For $0 \leq m < p$, differentiating the Fourier integral gives $\widehat{\psi}^{(m)}(0) = (-\mathrm{j})^m \int t^m \psi(t)\, \mathrm{d}t = 0$. The two-scale and conjugate-mirror identities give

$$\widehat{\psi}(2\omega) = -\frac{\mathrm{e}^{-\mathrm{j}N\omega}}{\sqrt{2}} \widehat{g}_0(\omega + \pi)^* \widehat{\phi}(\omega).$$

Since $\widehat{\phi}(0) = 1$, its factor is nonzero near zero. Thus the moments vanish precisely when $\widehat{g}_0$ has a zero of order at least p at π, or $G_0(z) = (1 + z^{-1})^p Q(z^{-1})$ for a real polynomial Q. The substitution $z^{-1} = \mathrm{e}^{-\mathrm{j}\omega}$ preserves the zero's order because its derivative is nonzero.

The squared magnitude of a real FIR filter is an even trigonometric polynomial. Pairing opposite lags gives cosines, and the recurrence $\cos((r+1)\omega) = 2\cos\omega\cos(r\omega) - \cos((r-1)\omega)$ expresses each cosine as a polynomial in $\cos\omega$. Hence

$$\tfrac{1}{2}|\widehat{g}_0(\omega)|^2 = R(\cos\omega), \qquad \deg R \leq N.$$

Power complementarity gives $R(x) + R(-x) = 1$. The factorization of G_0 implies that $R(x)$ contains $(1 + x)^p$, because

$$|1 + \mathrm{e}^{-\mathrm{j}\omega}|^{2p} = 2^p (1 + \cos\omega)^p.$$

Consequently $1 - R(x) = R(-x)$ contains $(1-x)^p$. The polynomial R' therefore has zeros of multiplicity at least $p-1$ at both endpoints. Since $R(-1) = 0$ and $R(1) = 1$, it is not identically zero, so

$$N - 1 \geq \deg R' \geq 2(p-1), \qquad L = N + 1 \geq 2p.$$

□

Why this also bounds the wavelet support. We measure support length as the length of the smallest interval containing the essential support, at scale zero. Keep the normalization $g_0[0]g_0[N] \neq 0$ and let $[a, b]$ be that interval for the compactly supported scaling function.

In

$$\phi(t) = \sqrt{2} \sum_{n=0}^{N} g_0[n]\phi(2t - n),$$

the leftmost possible point on the right is $a/2$, contributed by the $n = 0$ term. In a sufficiently small interval immediately to its right, all other terms vanish: their left endpoints are at least $(a+1)/2$. The nonzero endpoint tap cannot be canceled there. Thus the left endpoint of the right-hand side really is $a/2$. Equality of the two sides gives $a = a/2$, hence $a = 0$.

The same reasoning at the right edge gives $b = (b + N)/2$, hence $b = N$. This argument uses essential endpoints, so it does not require a nonzero value at a single endpoint.

For the conjugate-mirror filter $g_1[n] = (-1)^n g_0[N - n]$, both endpoint taps are nonzero. Since

$$\psi(t) = \sqrt{2} \sum_{n=0}^{N} g_1[n]\phi(2t - n),$$

the same uncancelled edge terms put the wavelet's support endpoints at 0 and N. Therefore

$$\boxed{\text{support length of } \phi = \text{support length of } \psi = L - 1 \geq 2p - 1.} \tag{12.11}$$

Other sign or translation conventions can move the support interval without changing its length. At scale i, its length is $2^{-i}(L-1)$.

Daubechies' 1988 construction attains both bounds: $2p$ filter taps and support span $2p - 1$. Thus, within this orthonormal setting, the Daubechies family has the shortest possible support and filters for a prescribed number of vanishing moments. We next construct a power response with the required endpoint zeros and factor it to obtain such a filter.

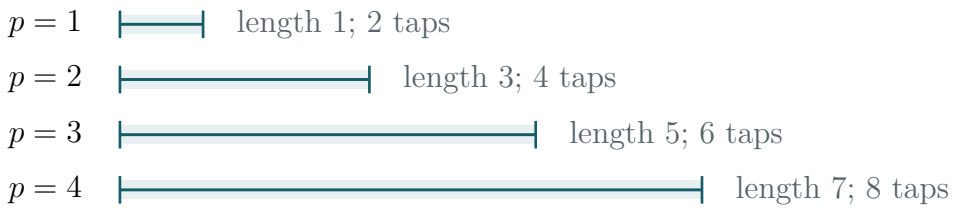


Figure 12.10: Minimum support spans at scale zero for the real orthonormal construction. More cancellation requires a longer footprint; dilation rescales every displayed length.

12.12 Why the minimum length can be attained

The orthogonality condition is expressed in terms of the squared magnitude of the filter response. We therefore design this power response first and then recover filter coefficients by factorization. This separates the constraints on magnitude from the subsequent choice of phase.

For $p \geq 1$, define

$$R_p(x) = c_p \int_{-1}^{x} (1-u^2)^{p-1}\,\mathrm{d}u, \qquad c_p = \left[\int_{-1}^{1} (1-u^2)^{p-1}\,\mathrm{d}u\right]^{-1}.$$

The integrand is an even polynomial of degree $2p-2$. Therefore R_p is a polynomial of degree $2p-1$ satisfying

$$R_p(-1) = 0, \qquad R_p(1) = 1, \qquad R_p(x) + R_p(-x) = 1.$$

The identity $R_p(x) + R_p(-x) = 1$ follows by substituting $u \mapsto -u$ in one integral and using the normalization of c_p. Near $x = -1$, the integrand has a zero of order $p-1$, so integration gives a zero of order exactly p. These two properties impose the complementary endpoint conditions used in the lower-bound proof.

Set

$$F_p(\omega) = 2R_p(\cos\omega). \tag{12.12}$$

Then $F_p \geq 0$, $F_p(0) = 2$, and

$$F_p(\omega) + F_p(\omega + \pi) = 2.$$

Also, F_p vanishes to order $2p$ at $\omega = \pi$, since $1 + \cos\omega$ vanishes to order two there. It is strictly positive everywhere else in a period.

For $p = 1$, $R_1(x) = (1+x)/2$, recovering the Haar power. For $p = 2$, integration gives $R_2(x) = 1/2 + 3x/4 - x^3/4$. The flatter endpoints improve cancellation, at the cost of a longer filter.

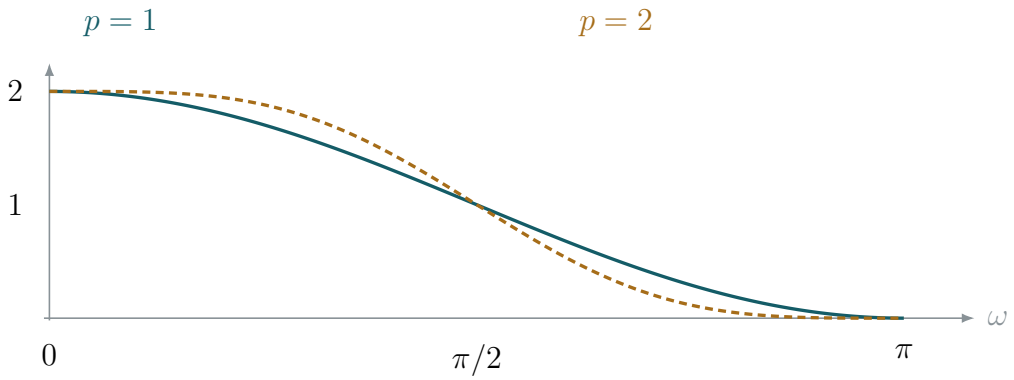


Figure 12.11: The first two minimum-length power responses. The two-moment design has flatter behavior at zero and π than Haar. The curves cross at the common half-band value one.

Recovering a real filter from its power. Put $N = 2p - 1$ and form the real Laurent polynomial

$$\mathcal{P}(z) = 2R_p\left(\frac{z + z^{-1}}{2}\right).$$

On the unit circle, $\mathcal{P}(e^{j\omega}) = F_p(\omega)$. We seek a real filter satisfying $\mathcal{P}(z) = G_0(z)G_0(z^{-1})$.

The ordinary polynomial $z^N\mathcal{P}(z)$ has degree $2N$, a nonzero constant term, and reciprocal real coefficients. Its roots therefore occur in reciprocal and conjugate pairs: if r is a root, then so are $1/r$, r^*, and $1/r^*$, with multiplicity. Its only unit-circle root is -1, of multiplicity $2p$. This follows from the positivity and zero order of F_p established above.

Assign p copies of -1 to the low-pass factor. The remaining roots come in reciprocal pairs off the unit circle. Taking the inside root from each pair gives $p - 1$ additional roots; because conjugate roots have the same magnitude, this selection is closed under conjugation. The degree-N polynomial

$$Q(z) = \prod_{r \text{ chosen}} (z - r)$$

has real coefficients. Both $z^{-N}Q(z)$ and its reciprocal partner together have exactly the zeros of $\mathcal{P}$, with the same multiplicities. Hence

$$Q(z)Q(z^{-1}) = c\,\mathcal{P}(z)$$

for a real constant c. At $z = 1$, both sides are positive, so $c > 0$. Rescaling Q and choosing its overall sign gives

$$G_0(z) = z^{-N}Q(z), \qquad G_0(1) = \sqrt{2}, \qquad G_0(z)G_0(z^{-1}) = \mathcal{P}(z).$$

This is a real FIR filter of length $N+1=2p$, with a zero of order p at $z=-1$ and the required power complementarity.

Finally, for $|\omega|\le\pi/2$, $\cos\omega\ge 0$, so $|\hat{g}_0(\omega)|^2 = 2R_p(\cos\omega)\ge 1$. The sufficient scaling-function theorem from Lecture 11 applies: the normalized infinite product yields an orthonormal MRA. Its conjugate-mirror wavelet has p vanishing moments and support length $2p-1$. This completes the attainment argument. Selecting the inside roots is the usual minimum-phase choice, apart from the required unit-circle zeros.

12.13 The four-tap, two-moment example

For $p=2$, the minimum filter length is four. The general construction can be seen directly by solving for four real taps (a,b,c,d). The DC normalization and the first zero at π give

$$a+b+c+d=\sqrt{2}, \qquad a-b+c-d=0.$$

The second vanishing moment imposes $-b+2c-3d=0$. The only nontrivial even-shift correlation is $ac+bd=0$.

Put $u=1/\sqrt{2}$. The first two equations give $a+c=b+d=u$. Substitution into the third gives $a+d=u/2$, hence

$$c=u-a, \qquad d=u/2-a, \qquad b=u/2+a.$$

The correlation condition becomes

$$a(u-a)+(u/2+a)(u/2-a)=0 \quad\Longleftrightarrow\quad 2a^2-ua-u^2/4=0.$$

Taking $a=u(1+\sqrt{3})/4$ produces

$$\boxed{g_0=\frac{1}{4\sqrt{2}}(1+\sqrt{3},\ 3+\sqrt{3},\ 3-\sqrt{3},\ 1-\sqrt{3}).} \tag{12.13}$$

The other root reverses the taps. We have not separately imposed unit norm: it follows from

$$a^2+b^2+c^2+d^2=(a+c)^2+(b+d)^2-2(ac+bd)=1.$$

All other nonzero even shifts have disjoint support. Thus the even translates are orthonormal. Its factorization is

$$G_0(z)=\frac{(1+z^{-1})^2}{4\sqrt{2}}\big[(1+\sqrt{3})+(1-\sqrt{3})z^{-1}\big].$$

The double zero is explicit. Moreover, its squared magnitude is $1 + \frac{3}{2}\cos\omega - \frac{1}{2}\cos^3\omega = 2R_2(\cos\omega)$, so the scaling-function existence condition checked above holds.

The high-pass filter is $g_1 = (d, -c, b, -a)$, and the analysis filters are $h_\ell[n] = g_\ell^*[-n]$. This wavelet cancels affine pieces, unlike Haar, and has support length three rather than one.

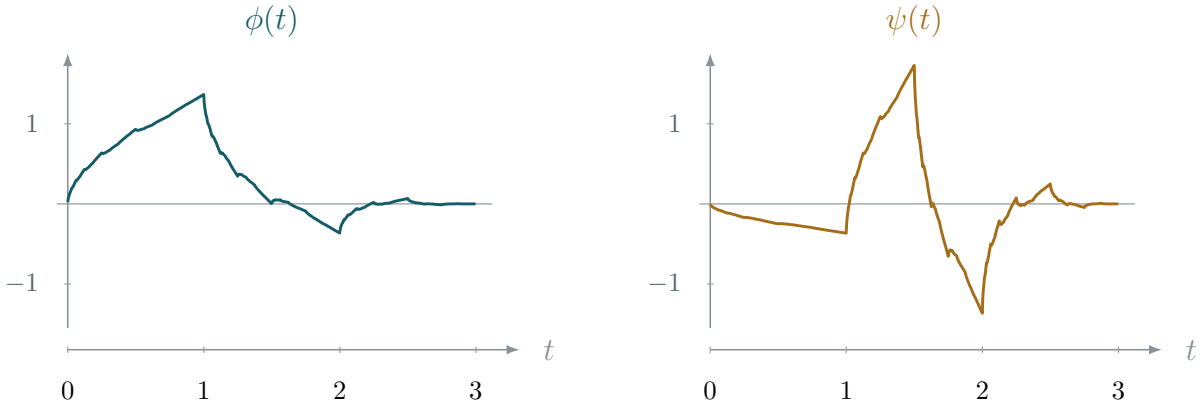


Figure 12.12: The four-tap scaling function and conjugate-mirror wavelet on a common vertical scale. The curves are numerical approximations from ten box-seeded low-pass refinements and one high-pass refinement. Both support spans are $[0, 3]$; the wavelet has two vanishing moments.

The design tradeoff. More vanishing moments improve cancellation on smooth pieces, but increase the minimum support and filter length. A longer support overlaps more breakpoints. If $\operatorname{supp}\psi \subseteq [0, L_\psi]$, the condition that $\psi_{i,n}$ straddle a breakpoint τ is

$$2^i\tau - L_\psi < n < 2^i\tau.$$

This interval contains at most $\lceil L_\psi \rceil$ integer indices. Thus S breakpoints can affect at most $S\lceil L_\psi \rceil$ wavelets at each scale. For piecewise polynomials of degree below p, every other detail coefficient vanishes by local polynomial cancellation.

Moments p	Minimum filter length	Minimum support length
1	2	1
2	4	3
3	6	5
4	8	7

Daubechies wavelets attain these bounds in the real orthonormal MRA setting. Symmlets use other phase choices to improve symmetry while retaining compact

support and vanishing moments. Orthogonalized spline wavelets need not be compactly supported; biorthogonal spline systems are outside the scope of this bound.

The examples show why a change of representation can expose structure that is not apparent from the signal samples. Wavelets combine octave organization with localization and polynomial cancellation. These properties favor signals that are simple on many local intervals, whereas a global sinusoid is more naturally represented by a small number of Fourier coefficients. Exact cancellation explains the piecewise-polynomial examples. For an approximately polynomial signal, the next question is how the local approximation error controls the coefficients that remain.